Daniel Haehn, University of Massachusetts Boston haehn@cs.umb.edu haehn danielhaehn.com

I am a biomedical imaging and visualization researcher who investigates how computational methods can accelerate biological and medical research.

## Education

| | | |
|---|---|---|
| 2019 | **PhD in Computer Science, Harvard University**<br>*Analyzing Brain Connectivity and Computing Machine Perception*<br>Advisor: Hanspeter Pfister<br>Committee: Steven Gortler, Finale Doshi-Velez, Scott Kuindersma, Jeff W. Lichtman | Cambridge, MA |
| 2010 | **Diplom (MSc) in Medical Computer Science, University of Heidelberg**<br>*Signal- and Image Processing*<br>Thesis: Coronary Artery Centerline Extraction<br>Advisors: Hartmut Dickhaus, Ron Kikinis | Germany |
| 2007 | **Vordiplom (BSc) in Medical Computer Science, University of Heidelberg**<br>*with Honors*, rank #1 of class, all study fees waived | Germany |

## Experience

| | | |
|---|---|---|
| 2019–present | **University of Massachusetts Boston**<br>*Associate Professor of Computer Science with Tenure (since 2025)*<br>*Director of the Machine Psychology research group*<br>*Director of the Artificial Intelligence (AI) Research Core Facility*<br>*Associate of Dana-Farber/Harvard Cancer Center*<br>*Associate of the Harvard John A. Paulson School of Engineering and Applied Sciences* | Boston, MA |
| Summer 2017 | **Apple, Inc.**<br>*Research Intern in Data Science* | Cupertino, CA |
| Summer 2014 | **Mental Canvas**<br>*Research Intern in Computer Graphics* | New York City, NY |

## Experience (continued)

2011–2013 **Boston Children's Hospital** Boston, MA
*Research Software Developer III*, Fetal Neonatal Neuroimaging and Developmental Science Center
Advisors: Rudolph Pienaar, P. Ellen Grant

2010–2011 **University of Pennsylvania** Philadelphia, PA
*Research Scholar*, Section for Biomedical Image Analysis
Advisor: Kilian Pohl

2009 **German Cancer Research Center (DKFZ) and BioQuant Center** Heidelberg, Germany
*Research Assistant*, Biomedical Computer Vision and Experimental Radiology Research Groups
Advisors: Stefan Wörz, Hendrik von Tengg-Kobligk

2008–2009 **Brigham and Women's Hospital** Boston, MA
*Fellow*, Department of Radiology and the Surgical Planning Laboratory
Advisors: Ron Kikinis, Steve Pieper, Luca Antiga

## Publications

**From UMass Boston** (* undergraduate student, ** graduate student; all peer-reviewed)

2026 **Avanith Kanamarlapudi, *Ryan Zurrin, *Edward Gaibor, **Benjamin Bendiksen Gutierrez, **Neha Goyal, **Vidhya Sree Narayanapa, Dan Simovici, Nurit Haspel, Marc Pomplun, Hyunkwang Lee, Mack Bandler, Greg Sorensen, and Daniel Haehn. **OMAMA-DB: the Oregon-Massachusetts Mammography Database.** *Journal of Medical Imaging*.

2026 **Jiehyun Kim, Huy Q Phi, Grace J. Wang, Brett L. Cucchiara, Elias Johansson, Jae W. Song, and Daniel Haehn. **Uncertainty Quantification from Calcified Plaque to Outer Wall Estimation in CTA.** *Medical Imaging with Deep Learning*.

2026 *Moody, Blake, **Kim, JieHyun, **Kim, Sanghyuk, and Daniel Haehn. **Machine learning for N-dimensional spatial reasoning tasks on the web.** *Frontiers in Bioinformatics*.

2026 **Jiang, Shuning, Wei-Lun Chao, Daniel Haehn, Hanspeter Pfister, and Jian Chen. **A Rigorous Behavior Assessment of CNNs Using a Data-Domain Sampling Regime.** *IEEE Transactions on Visualization and Computer Graphics*.

2026 Yu Sakai, **Kim, Jiehyun, Huy Q. Phi, Andrew C. Hu, Pargol Balali, Konstanze V. Guggenberger, John H. Woo, Daniel Bos, Scott E. Kasner, Brett L. Cucchiara, Luca Saba, Zhi Huang, Daniel Haehn, and Jae W. Song. **Explainable Machine-Learning Model to Classify Culprit Calcified Carotid Plaque in ESUS.** *Journal of Neuroimaging*.

2025 **Kyeremah, Charlotte, Aditya S Paul, Daniel Haehn, Manoj T Duraisingh, and Chandra S Yelleswarapu. **Enhanced detection of malaria infected red blood cells through phase driven classification.** *Scientific Reports*.

2025 Arianna Fernandez*, Sophia de Oliveira*, Chaya Spencer*, Kalysha Melendez Hernandez*, Samatrai Piam*, Daniel Haehn, and Amanda Potasznik. **CTRL + Empower: Lived Experiences of Black Women and Latinas in Computer Science at an Urban Public University**. *IEEE International Symposium on Ethics in Science, Technology and Engineering (ETHICS)*.

2025 Rami Huu Nguyen**, Kenichi Maeda*, Mahsa Geshvadi**, and Daniel Haehn. **Evaluating 'Graphical Perception' with Multimodal LLMs.** *IEEE Pacific Visualization (PacificVis)*.

2025 SangHyuk Kim**, Edward Gaibor*, Brian Matejek, and Daniel Haehn. **Melanoma Detection with Uncertainty Quantification**. *IEEE International Symposium on Biomedical Imaging (ISBI)*.

2025 Edward Gaibor*, Shruti Varade**, Rohini Deshmukh**, Tim Meyer**, Mahsa Geshvadi**, SangHyuk Kim**, Vidhya Sree Narayanappa**, and Daniel Haehn. **Boostlet.js: Medical Image Processing Plugins for the Web via JavaScript Injection**. *IEEE International Symposium on Biomedical Imaging (ISBI)*.

2025 Jiehyun Kim**, Kevin Wang, Yu Sakai, Youxiang Zhu, Andrew C. Hu, Huy Q. Phi, Nathan Arnett, Grace J. Wang, Brett L. Cicchiara, Jae W. Song, and Daniel Haehn. **Automatic Segmentation of Calcified Plaque in Carotid Arteries**. *IEEE International Symposium on Biomedical Imaging (ISBI)*.

2024 Youxiang Zhu**, Nana Lin**, Kiran Sandilya Balivada**, Daniel Haehn, and Xiaohui Liang. **Adversarial Text Generation using Large Language Models for Dementia Detection**. *Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

2024 Tim Meyer**, Gabi Dreo Rodosek, and Daniel Haehn. **WebGL-based Image Processing through JavaScript Injection**. *ACM Conference on 3D Web Technology*.

2024 Zhenxing Cui, Lu Chen, Yunhai Wang, Daniel Haehn, Yong Wang, and Hanspeter Pfister. **Generalization of CNNs on Relational Reasoning With Bar Charts.** *IEEE Transactions on Visualization and Computer Graphics*.

2024 Amanda R. Potasznik, Funda Durupinar, and Daniel Haehn. **'Just let me': Experiences of Historically Marginalized Students in the CS Department**. *IEEE Black Issues in Comp. Education (BICE)*.

2024 Loraine Franke**, Tae Young Park, Jie Luo, Yogesh Rathi, Steve Pieper, Lipeng Ning, and Daniel Haehn. **SlicerTMS: Real-Time Visualization of Transcranial Magnetic Stimulation for Mental Health Treatment**. *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.

2024 Charlotte Kyeremah**, Matthew Weiss**, Dila Kandel**, Daniel Haehn, and Chandra Yelleswarapu. **Single-beam digital holographic reconstruction: a phase-support enhanced complex wavefront on phase-only function for twin-image elimination.** *Journal of Biomedical Optics*.

2024 Loraine Franke**, Daniel Karl I. Weidele, Nima Dehmamy, Lipeng Ning, and Daniel Haehn. **AutoRL X: Automated Reinforcement Learning on the Web.** *ACM Transactions Interactive Intelligent Systems*.

2024 Akshata Tiwari*, Brian Matejek, and Daniel Haehn. **Non-invasive Stress Monitoring from Video**. *IEEE International Symposium on Biomedical Imaging (ISBI)*.

2024 Tae Young Park, Loraine Franke**, Steve Pieper, Daniel Haehn, and Lipeng Ning. **A review of algorithms and software for real-time electric field modeling techniques for transcranial magnetic stimulation.** *Biomedical Engineering Letters*.

2023 Navaneethakrishna Makaram, Sarvagya Gupta**, Matthew Pesce, Jeffrey Bolton, Scellig Stone, Daniel Haehn, Marc Pomplun, Christos Papadelis, Phillip Pearl, Alexander Rotenberg, Patricia Ellen Grant, and Eleonora Tamilia. **Deep Learning-Based Visual Complexity Analysis of Electroencephalography Time-Frequency Images: Can It Localize the Epileptogenic Zone in the Brain?**. *MDPI Algorithms*.

2023 Kristin Qi**, Jiali Cheng, and Daniel Haehn. **Lesion Search with Self-supervised Learning.** *International Conference on Learning Representations (ICLR)*.

2023 Ryan Zurrin*, Neha Goyal**, Pablo Bendiksen**, Muskaan Manocha**, Dan Simovici, Nurit Haspel, Marc Pomplun, and Daniel Haehn. **Outlier Detection for Mammograms**. *International Conference on Medical Imaging with Deep Learning (MIDL)*.

2023 Ramin Dehghanpoor**, Fatemeh Afrasiabi**, Charles Fogel*, Tung Dao*, Suman Gautam*, Aanab Nehela*, Ahmad Nehela*, Daniel Haehn, and Nurit Haspel. **Classifying Protein Families with Learned Compressed Representations**. *International Conference on Bioinformatics (Best Paper Award at BICOB)*.

2023 Daniel Karl I. Weidele, Shazia Afzal, Abel N. Valente, Cole Makuch, Owen Cornec, Long Vu, Dharmashankar Subramanian, Werner Geyer, Rahul Nair, Inge Vejsbjerg, Radu Marinescu, Paulito Palmes, Elizabeth M. Daly, Loraine Franke**, and Daniel Haehn. **AutoDOViz: Human-Centered Automation for Decision Optimization**. *ACM International Conference on Intelligent User Interfaces (IUI)*.

2022 Neha Goyal**, Yahiya Hussain*, Gianna G. Yang*, and Daniel Haehn. **Real-Time Alignment for Connectomics**. *Springer LNCS: Biomedical Image Registration (WBIR)*.

2022 Francois Rheault, Valérie Hayot-Sasson, Robert E. Smith, Christopher Rorden, Jacques-Donald Tournier, Eleftherios Garyfallidis, Fang-Cheng Yeh, Christopher J. Markiewicz, Matthew Brett, Ben Jeurissen, Paul A. Taylor, D. Baran Aydogan, Derek A. Pisner, Serge Koudoro, Soichi Hayashi, Daniel Haehn, Steve Pieper, Daniel Bullock, Emanuele Olivetti, Jean-Christophe Houde, Marc-Alexandre Côté, Flavio Dell'Acqua, Alexander Leemans, Maxime Descoteaux, Bennett Landman, Franco Pestilli, and Ariel Rokem. **TRX: A community-oriented tractography file format**. *OHBM 2022-Human Brain Mapping (Oral)*.

2022 Jay Burkhardt*, Aaryaman Sharma*, Jack Tan*, Loraine Franke**, Jahnavi Leburu**, Jay Jeschke, Sasha Devore, Daniel Friedman, Jingyun Chen, and Daniel Haehn. **N-Tools-Browser: Web-Based Visualization of Electrocorticography Data for Epilepsy Surgery**. *Frontiers in Bioinformatics*.

2022 Katharina Paulick, Simon Seidel, Christoph Lange, Annina Kemmer, Mariano Nicolas Cruz-Bournazou, André Baier, and Daniel Haehn. **Promoting Sustainability through Next-Generation Biologics Drug Development**. *MDPI Sustainability*.

2022 Nalini M. Singh, Jordan B. Harrod, Sandya Subramanian, Mitchell Robinson, Ken Chang, Suheyla Cetin-Karayumak, Adrian Vasile Dalca, Simon Eickhoff, Michael Fox, Loraine Franke**, Polina Golland, Daniel Haehn, Juan Eugenio Iglesias, Lauren J. O'Donnell, Yangming Ou, Yogesh Rathi, Shan H. Siddiqi, Haoqi Sun, M. Brandon Westover, Susan Whitfield-Gabrieli, and Randy L. Gollub. **How Machine Learning is Powering Neuroimaging to Improve Brain Health**. *Neuroinformatics*.

2021 Bella Baidak*, Yahiya Hussain*, Emma Kelminson*, Thouis R. Jones, Loraine Franke**, and Daniel Haehn. **CellProfiler Analyst Web (CPAW) - Exploration, analysis, and classification of biological images on the web**. *IEEE Visualization Short Paper (IEEE VIS)*.

2021 Loraine Franke**, Daniel Karl I Weidele, Fan Zhang, Suheyla Cetin-Karayumak, Steve Pieper, Lauren J O'Donnell, Yogesh Rathi, and Daniel Haehn. **FiberStars: Visual Comparison of Diffusion Tractography Data between Multiple Subjects**. *IEEE Pacific Visualization (PacificVis)*.

2020 Loraine Franke** and Daniel Haehn. **Modern Scientific Visualizations on the Web**. *MDPI Informatics*.

2020 Daniel Haehn, Loraine Franke**, Fan Zhang, Suheyla Cetin Karayumak, Steve Pieper, Lauren O'Donnell, and Yogesh Rathi. **TRAKO: Efficient Transmission of Tractography Data for Visualization**. *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.

2020 Vincent Casser, Kai Kang, Hanspeter Pfister, and Daniel Haehn. **Fast Mitochondria Detection for Connectomics**. *International Conference on Medical Imaging with Deep Learning (Spotlight Award at MIDL)*.

2020 Zudi Lin, Donglai Wei, Won-Dong Jang, Siyan Zhou, Xupeng Chen, Xueying Wang, Richard L. Schalek, Daniel R. Berger, Brian Matejek, Lee D. Kamentsky, Adi Peleg, Daniel Haehn, Thouis R. Jones, Toufiq Parag, Jeff W. Lichtman, and Hanspeter Pfister. **Two-Stream Active Query Suggestion for Large-Scale Object Detection in Connectomics**. *European Conference on Computer Vision (ECCV)*.

2020 Fritz Lekschas, Brant Peterson, Daniel Haehn, Eric Ma, Nils Gehlenborg, and Hanspeter Pfister. **Peax: Interactive Visual Pattern Search in Sequential Data Using Unsupervised Deep Representation Learning**. *Computer Graphics Forum (Best Paper Award at EuroVis)*.

## Prior to UMass Boston

2019 Brian Matejek, Daniel Haehn, Haidong Zhu, Donglai Wei, Toufiq Parag, and Hanspeter Pfister. **Biologically-Constrained Graphs for Global Connectomics Reconstruction**. *IEEE Computer Vision and Pattern Recognition (CVPR)*.

2018 Daniel Haehn, James Tompkin, and Hanspeter Pfister. **Evaluating 'Graphical Perception' with CNNs**. *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS)*.

2018 Daniel Haehn, Verena Kaynig, James Tompkin, Jeff W. Lichtman, and Hanspeter Pfister. **Guided Proofreading of Automatic Segmentations for Connectomics**. *IEEE Computer Vision and Pattern Recognition (CVPR)*.

2017 Daniel Haehn, John Hoffer, Brian Matejek, Adi Suissa-Peleg, Ali K. Al-Awami, Lee Kamentsky, Felix Gonda, Eagon Meng, William Zhang, Richard Schalek, Alyssa Wilson, Toufiq Parag, Johanna Beyer, Verena Kaynig, Thouis R. Jones, James Tompkin, Markus Hadwiger, Jeff W. Lichtman, and Hanspeter Pfister. **Scalable Interactive Visualization for Connectomics**. *MDPI Informatics*.

2017 Brian Matejek, Daniel Haehn, Fritz Lekschas, Michael Mitzenmacher, and Hanspeter Pfister. **Compresso: Efficient Compression of Segmentation Data For Connectomics**. *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.

2017 Felix Gonda, Verena Kaynig, Thouis R. Jones, Daniel Haehn, Jeff W. Lichtman, Toufiq Parag, and Hanspeter Pfister. **ICON: An Interactive Approach to train Deep Neural Networks for Segmentation of Neuronal Structures**. *IEEE International Symposium on Biomedical Imaging (ISBI)*.

2017 Rudolph Pienaar, Ata Turk, Jorge Bernal-Rusiel, Nicolas Rannou, Daniel Haehn, P. Ellen Grant, and Orran Krieger. **CHIPS--A Service for Collecting, Organizing, Processing, and Sharing Medical Image Data in the Cloud**. *VLDB Workshop on Data Management and Analytics for Medicine and Healthcare*.

2016 Adi Suissa-Peleg, Daniel Haehn, Seymour Knowles-Barley, Verena Kaynig, Thouis R. Jones, Alyssa Wilson, Richard Schalek, Jeff W. Lichtman, and Hanspeter Pfister. **Automatic Neural Reconstruction from Petavoxel of Electron Microscopy Data**. *Microscopy and Microanalysis*.

2016 Ali K. Al-Awami, Johanna Beyer, Daniel Haehn, Narayanan Kasthuri, Jeff W. Lichtman, Hanspeter Pfister, and Markus Hadwiger. **NeuroBlocks--Visual Tracking of Segmentation and Proofreading for Large Connectomics Projects**. *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS)*.

2016 Richard Schalek, Dong Lee, Narayanan Kasthuri, Adi Peleg, Thouis R. Jones, Verena Kaynig, Daniel Haehn, Hanspeter Pfister, David Cox, and Jeff W. Lichtman. **Imaging a 1 mm$^3$ Volume of Rat Cortex using a Multi-Beam SEM**. *Microscopy and Microanalysis*.

2015 Kiho Im, Banu Ahtam, Daniel Haehn, Jurriaan M. Peters, Simon K. Warfield, Mustafa Sahin, and P. Ellen Grant. **Altered Structural Brain Networks in Tuberous Sclerosis Complex**. *Cerebral Cortex*.

2015 Rudolph Pienaar, Nicolas Rannou, Jorge Bernal, Daniel Haehn, and P. Ellen Grant. **ChRIS--A web-based Neuroimaging and Informatics System for Collecting, Organizing, Processing, Visualizing and Sharing of Medical Data**. *IEEE Engineering in Medicine and Biology Society (EMBC)*.

2014 Daniel Haehn, Seymour Knowles-Barley, Mike Roberts, Johanna Beyer, Narayanan Kasthuri, Jeff W. Lichtman, and Hanspeter Pfister. **Design and Evaluation of Interactive Proofreading Tools for Connectomics.** *IEEE Transactions on Visualization and Computer Graphics (IEEE VIS)*.

2013 Daniel Haehn, Nicolas Rannou, P. Ellen Grant, and Rudolph Pienaar. **Slice:Drop -- Collaborative Medical Imaging in the Browser.** *ACM SIGGRAPH Computer Animation Festival*.

2012 Daniel Haehn, Nicolas Rannou, Banu Ahtam, P. Ellen Grant, and Rudolph Pienaar. **Neuroimaging in the Browser using the X Toolkit.** *Frontiers in Neuroinformatics (Spotlight Award at INCF Neuroinformatics)*.

2012 Myong-sun Choe, Silvia Ortiz-Mantilla, Nikos Makris, Matt Gregas, Janine Bacic, Daniel Haehn, David Kennedy, Rudolph Pienaar, Verne S. Caviness Jr, April A. Benasich, and P. Ellen Grant. **Regional Infant Brain Development: an MRI-based Morphometric Analysis in 3 to 13 month olds.** *Cerebral Cortex*.

2012 Arno Klein, Forrest S. Bao, Yrjö Häme, Eliezer Stavsky, Joachim Giard, Daniel Haehn, Nolan Nichols, and Satrajit S. Ghosh. **Mindboggle: Automated Human Brain MRI Feature Extraction, Labeling, Morphometry, and Online Visualization.** *Frontiers in Neuroinformatics*.

2012 Arno Klein, Nolan Nichols, and Daniel Haehn. **Mindboggle 2 interface: Online Visualization of Extracted Brain Features with XTK.** *Frontiers in Neuroinformatics*.

## Grants

**Funded** (* Principal Investigator)

2026–2029 *Massachusetts Life Sciences Center, Bits to Bytes: **SPLEEN-US: Open Annotated Spleen Ultrasound Dataset for Precision Neuromodulation**, (Co-PIs: Dragana Savic, Tracy Baynard, Bo Fernhall), $745,400.00

2026 *National Science Foundation, ACCESS: **NeurodeskEDU: Scalable, Hands-On Training for Modern Neuroimaging Using ACCESS Resources**, (Co-PIs: Chris Rorden, David Smith, Dianne Patterson, Andrew Jahn, Steffen Bollmann), $25,000 in compute credits

2025–2027 *Sloan Foundation Grant: **Expanding Culture Change across STEM Disciplines at the University of Massachusetts Public Education System: A Partnership Between UMass Boston (UMB) and Amherst (UMA)**, (Co-PIs UMB: Kimberly Hamad Schifferli, UMA: Neena Thota, Shannon Roberts), $500,000.00

2023–2025 National Institutes of Health, U54 Supplement: **iTCGA---Integrated Training in Computational Genomics and Data Sciences**, Co-PI (PI Adan Colon-Carmona, PI Jill Mascoska, Co-PI Brooke Moyers, Co-PI Kourosh Zarringhalam, Co-PI Christina Kelliher; UMB), $581,237.00

## Grants (continued)

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| 2022–2024 | Sloan Foundation Grant: **Culture Change in Computer Science and Engineering at the University of Massachusetts Public University System: A Partnership Between UMass Boston and Amherst**, UMB Site-PI (Co-PI Hamad Schifferli, UMA: PI Dasgupta, Co-PI Thota, Co-PI Roberts), $499,972.00 (UMB share $238,948.00) |
| 2021–2023 | National Institutes of Health, R21: **Real-time visualization and precision targeting in transcranial magnetic stimulation**, Co-PI (PI Lipeng Ning, Harvard Medical School), $493,011.00 (UMB share $156,663.00) |
| 2021–2022 | UMass Boston, Proposal Development Grant: **Towards Developing Deep Learning Approaches for Protein-Protein Interaction Detection**, Co-PI (PI Nurit Haspel, UMB), $20,000.00 |
| 2020–2023 | *Massachusetts Life Sciences Center, Bits to Bytes: **The Oregon-Massachusetts Mammography Database (OMAMA-DB)**, (Co-PIs Haspel, Tonyushkin, Pomplun, Simovici), $749,834.00 |
| 2020 | Federal Ministry of Education and Research Germany: International Future Labs for Artificial Intelligence in collaboration with the KIWI Biolab at the Technical University Berlin (covering 18 months exchange visits of a PostDoc and a Ph.D. student), (PI Cruz-Bournazou, TU Berlin) |
| 2019 | *NVidia Accelerated Data Science GPU Grant, 1x Titan V100 GPU |

## Presentations

* invited presentation

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|---|---|
| 2026 | ***Fetal Neonatal Neuroimaging Developmental Science Center** at Boston Children's Hospital |
| 2026 | *Co-Organizer and Presenter at the **A.I. Summit** at UMass Boston |
| 2025 | Speaker at the **Sloan Faculty Learning Community**: *Difficult Conversations* |
| 2025 | *Presenter to the **AI Delegation of India** at the Venture Development Center, UMass Boston |
| 2025 | *Organizer and Presenter for the **Lab Central Tour** at UMass Boston |
| 2025 | *Presenter to the Delegation of **Massachusetts Secretary of Labor & Worforce Development Lauren Jones** |
| 2025 | *Presenter to the **AI Delegation of Georgia (Europe)** at the Venture Development Center, UMass Boston |
| 2025 | *Speaker at **Visualizing Biological Data (VIZBI)**: *Masterclass on Web Graphics*, University of Cambridge, UK |
| 2025 | *Panelist for **Employment Opportunities in the Age of Cyber Technology Proliferation and AI** at Brandeis University |
| 2025 | ***AI Core Capabilities at the Venture Development Center** at UMass Boston |
| 2025 | **National Research Mentoring Network** Instructor for Graduate Students at UMass Boston |
| 2024 | ***High-Performance Computing and AI Core Presentation** for the Healey Library at UMass Boston |
| 2024 | Speaker at the **Sloan Faculty Learning Community**: *Difficult Conversations* |
| 2024 | ***Center for Innovative Teaching Presentation of Inclusive Pedagogies** at UMass Boston |
| 2024 | **National Research Mentoring Network** Instructor for Faculty at UMass Boston |

## Presentations (continued)

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|---|---|
| 2024 | *Panelist at **TEACH**: *Creative Assignments in the age of AI* |
| 2023 | Speaker at the **Sloan Faculty Learning Community**: *Difficult Conversations* |
| 2023 | *Speaker at **MIT Brainhack**: *TRAKO and beyond* |
| 2023 | *Speaker at the **Hack NiiVue.js!, University of South Carolina**: *BOOSTLET.js* |
| 2022 | *Speaker at the **High Performance Computing Day**: *Processing of Massive Biological Datasets at UMB* |
| 2022 | *Speaker at **Visualizing Biological Data (VIZBI)**: *Masterclass on Scientific Visualization* |
| 2021 | *Presenter at the **UMass Summit for AI, Data Science, and Robotics** |
| 2020 | *Presenter at the **Creative Commons Global Summit**: *The 7 Levels of Open Science* |
| 2020 | Presenter at the **National Alliance for Medical Image Computing Project Week**: *Integrating TRAKO with 3D Slicer* |
| 2020 | *Speaker at the **Fetal Neonatal Developmental Science Center, Boston Children's Hospital**: *Scientific Visualization at Scale!* |
| 2020 | Paper presentation at **International Conference on Medical Image Computing and Computer Assisted Intervention**: *TRAKO: Efficient Transmission of Tractography Data for Visualization* |
| 2020 | *Speaker at the **Lymph Node Quantification Project, Harvard Medical School**: *Machine-Guided Annotation Methods* |
| 2020 | Paper presentation at **Medical Imaging with Deep Learning (MIDL)**: *Fast Mitochondria Detection for Connectomics* |
| 2020 | *Presentation at the **UMass Boston-Dana Farber/Harvard Cancer Center initiative**: *Guided Tumor Detection and Annotation Methods for Cancer Imaging* |
| 2020 | *Speaker at the **Massachusetts Life Sciences Center**: *The Oregon-Massachusetts Mammography Database* |
| 2020 | *Researcher at **Shonan Meeting No. 167 in Japan**: *Formalizing Biological and Medical Visualization* |
| 2019 | *Speaker at **Sarah Frisken's Lab, Harvard Medical School**: *Brain Connectivity, Machine Perception, and Computer Graphics - all at different scales!* |
| 2019 | *Speaker at **Suffolk University**: *Brain Connectivity and Machine Perception* |
| 2019 | *Speaker at the **MIT McGovern Institute**: *The Performance Gap between the Brain and AI* |
| 2018 | Paper presentation at **IEEE Visualization**: *Evaluating 'Graphical Perception' with CNNs* |
| 2018 | **Harvard Visual Computing Group** meeting presentation: *The 7 Levels of Open Science* |
| 2018 | *Speaker at **Brown University, Department of Computer Science**: *Analyzing Brain Connectivity and Computing Machine Perception* |
| 2018 | *Speaker at **IBM Research** (AI Systems Day): *Evaluating 'Graphical Perception' with CNNs* |
| 2017 | **Harvard Visual Computing Group** meeting presentation: *Guided Proofreading of Automatic Segmentations for Connectomics* |
| 2016 | *Speaker at the **IEEE Visualization Doctoral Colloquium**: *Proofreading for Connectomics* |
| 2015 | **Harvard Lichtman Lab** meeting presentation: *Interactive Proofreading Tools for Connectomics* |

## Presentations (continued)

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|---|---|
| 2014 | Paper presentation at **IEEE Visualization**: *Design and Evaluation of Interactive Proofreading Tools for Connectomics* |
| 2014 | **Harvard Visual Computing Group** meeting presentation: *Proofreading Tools for Connectomics* |
| 2014 | *Speaker at the **MIT Computer Graphics Group**: *Web-based Visualization of Scientific Data* |
| 2014 | **Harvard Visual Computing Group** meeting presentation: *Interactive Proofreading with Dojo* |
| 2014 | **Harvard Lichtman Lab** meeting presentation: *Web-based Visualization and Proofreading for Connectomics* |
| 2013 | **Harvard Visual Computing Group** meeting presentation: *Web-based Scientific Visualization* |
| 2013 | *Speaker at **Visualizing Biological Data (VIZBI)**: *Physiology & Function* |
| 2012 | Spotlight presentation at **INCF Neuroinfomatics**: *Neuroimaging in the Browser using the X Toolkit* |
| 2012 | *Speaker at **WebGL Camp Orlando**: *WebGL for Baby Brains* |

## Awards

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|---|---|
| 2024 | Wasabi Fenway Bowl, Educator Shoutout at Fenway Park |
| 2024 | Excellence in Undergraduate Education Award in the College of Science and Mathematics at UMass Boston |
| 2024 | Center for Innovative Teaching (CIT) Award for Inclusive Pedagogies at UMass Boston |
| 2023 | Early Career Research Excellence Award in the College of Science and Mathematics at UMass Boston |
| 2023 | Best Paper Award at BICOB for Classifying Protein Families |
| 2022 | Selected for Oral Presentation at ISMRM Neuromodulation for TMS Visualization |
| 2022 | Selected for Oral Presentation at the Organization for Human Brain Mapping for TRX |
| 2020 | Best Paper Award at EuroVis for Peax |
| 2020 | Spotlight Award at MIDL: Fast Mitochondria Detection for Connectomics |
| 2020 | AI Scientist of the Future for the KIWI Biolab at the Technical University Berlin, Germany |
| 2015–2019 | Winkler Scholarship |
| 2013–2019 | Harvard University Fellowship |
| 2013 | Real-Time Live! presentation of Slice:Drop at SIGGRAPH |
| 2012 | INCF Neuroinformatics Spotlight Award for XTK |
| 2012 | Mozilla Hacks WebGL Dev Derby Runner-up for Slice:Drop |
| 2012 | Visualizing.org VisWeek Challenge Winner with Slice:Drop |
| 2010 | 1st Prize for End User Tutorial at the National Alliance of Medical Image Computing (NA-MIC) |
| 2008–2009 | Karl Steinbuch Foundation Scholarship |
| 2007–2009 | Thomas Gessmann Foundation Scholarship |

# Teaching

At UMass Boston (main instructor unless indicated, * re-designed course, ** new course)

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|---|---|
| 2026 | IMPACT 16-week AI Curriculum (20 students) |
| 2026 | **CS480/CS697 Special Topics: Visualizing.Boston (24 students, Instructor rating: 4.91/5, Course rating: 4.826/5) |
| 2026 | CS666 Biomedical Signal and Image Processing (18 students, Instructor rating: 4.67/5, Course rating: 4.44/5) |
| 2025 | CS460 Graphics (49 students, Instructor rating: 4.9/5, Course rating: 4.8/5) |
| 2024 | CS460 Graphics (40 students, Instructor rating: 4.93/5, Course rating: 4.9/5) |
| 2024 | **CS480/CS697 Special Topics: Visualizing.Boston (20 students, Instructor rating: 5/5, Course rating: 5/5) |
| 2024 | CS666 Biomedical Signal and Image Processing (19 students, Instructor rating: 4.68/5, Course rating: 4.63/5) |
| 2023 | CS460 Graphics (48 students, Instructor rating: 4.89/5, Course rating: 4.85/5) |
| 2023 | **CS666 Biomedical Signal and Image Processing (57 students, Instructor rating: 4.92/5, Course rating: 4.82/5) |
| 2023 | CS410 Introduction to Software Engineering (60 students, Instructor rating: 4.67/5, Course rating: 4.44/5) |
| 2023 | Guest Lecturer for CS615 User Interface Design |
| 2023 | Guest Lecturer at the Integrative Biosciences & Computational Sciences Graduate Program Seminar |
| 2022 | CS460 Graphics (48 students, Instructor rating: 4.76/5, Course rating: 4.79/5) |
| 2022 | Guest Lecturer for CS615 User Interface Design |
| 2022 | CS480/CS697 Special Topics: Biomedical Signal and Image Processing (27 students, Instructor rating: 5/5, Course rating: 4.93/5) |
| 2022 | CS410 Introduction to Software Engineering (62 students, Instructor rating: 4.23/5, Course rating: 4.41/5) |
| 2022 | Guest Lecturer for BIOL693 Seminar in Neurobiology |
| 2021 | Guest Lecturer for CS615 User Interface Design |
| 2021 | CS460 Graphics (20 students, Instructor rating: 5/5, Course rating: 4.94/5) |
| 2021 | **CS480/CS697 Special Topics: Biomedical Signal and Image Processing (27 students, Instructor rating: 4.8/5, Course rating: 4.75/5) |
| 2021 | CS410 Introduction to Software Engineering (47 students, Instructor rating: 4.59/5, Course rating: 4.45/5) |
| 2020 | CS460 Graphics (28 students, Instructor rating: 4.9/5, Course rating: 4.9/5) |
| 2020 | *CS410 Introduction to Software Engineering (27 students, Instructor rating: 4.87/5, Course rating: 4.73/5) |
| 2019 | **CS460 Graphics (24 students, Instructor rating: 4.81/5, Course rating: 4.57/5) |
| 2019 | Guest Lecturer for two lectures of the CS187 Science Gateway Seminar |

Outside UMass Boston

| | |
|---|---|
| 2021 | Guest Lecturer for CSCI2254 Web Application Development at Boston College |
| 2020 | Guest Lecturer for CSCI2254 Web Application Development at Boston College |
| 2019 | Guest Lecturer for the CMPSC131 Computer Science course at Suffolk University |
| 2018–2019 | TEALS Volunteer for AP Computer Science at Cambridge Rindge and Latin School |
| 2016 | Technical Assistant for the Deep Learning mini-course at the Harvard IACS Compute Fest |
| 2015 | Teaching Fellow for the Harvard CS171 Visualization course |
| 2008 | Workshop for Advanced Microcontroller Programming, University of Bratislava, Slovakia |
| 2008 | Workshop for Microcontroller Programming at the University of Tbilisi, Georgia (Europe) |
| 2004–2008 | Teaching Assistant for the Microcontrollers in EXperiment and LEarning (MEXLE) educational platform, Heilbronn University, Germany |

# Mentoring

Graduate Students (PhD)

| | |
|---|---|
| 2023–present | SangHyuk Kim, Computer Science, University of Massachusetts Boston |
| 2025–present | Avanith Kanamarlapudi, Computer Science, University of Massachusetts Boston |
| 2024–2026 | Rami Huu Nguyen, Computational Sciences, University of Massachusetts Boston |
| 2022–2024 | Mahsa Geshvadi, Computer Science, University of Massachusetts Boston |
| 2022–2026 | Jenna (JieHyun) Kim, Computer Science, University of Massachusetts Boston, *defended 4/2026* |
| 2020–2023 | Kristin (Yanan) Qi, Computational Sciences, University of Massachusetts Boston |
| 2021 | Hayoun Oh, Computer Science, Harvard University (co-mentored) |
| 2020–2021 | Aswin Vasudevan, Computer Science, University of Massachusetts Boston |
| 2019–2024 | Loraine Franke, Computer Science, University of Massachusetts Boston, *defended 4/2024* |
| 2019–2021 | Jesse Freeman, Computer Science, University of Massachusetts Boston |

Graduate Students (MS)

| | |
|---|---|
| 2023–2025 | Shruti Shailendra Varade, Computer Science, University of Massachusetts Boston |
| 2023–2025 | Vidhya Sree Narayanappa, Computer Science, University of Massachusetts Boston |
| 2022–2024 | Dhruv Shah, Computer Science, University of Massachusetts Boston |
| 2022–2023 | Kunal Jain, Computer Science, University of Massachusetts Boston |
| 2022–2025 | Kiran Sandilya, Computer Science, University of Massachusetts Boston |
| 2022 | Benni Bendiksen, Computer Science, University of Massachusetts Boston |
| 2021–2023 | Neha Goyal, Computer Science, University of Massachusetts Boston |

# Mentoring (continued)

| | |
|---|---|
| 2020 | Jiali Cheng, Computer Science, Northeastern University |
| 2020 | Gianna Yang, Computer Science, University of Massachusetts Boston |
| 2020 | Barkha Java, Computer Science, University of Massachusetts Boston |
| 2019 | Manish Mourya, Computer Science, University of Massachusetts Boston |
| 2018–2020 | Vincent Casser, Computer Science, Harvard University |
| 2010-2011 | Suares Tamekue, Intern at Brigham and Women's Hospital (co-mentored) |

### Undergraduate and Pre-College Students

| | |
|---|---|
| 2026–present | Ayden Bronsdon, Computer Science, University of Massachusetts Boston |
| 2026–present | Jonathan Allen, Computer Science, University of Massachusetts Boston |
| 2025–present | Mckenshel Cameau, Computer Science, University of Massachusetts Boston |
| 2024–2026 | Blake Moody, Computer Science, University of Massachusetts Boston |
| 2024–present | Daniel Gross, Computer Science, University of Massachusetts Boston |
| 2024–2025 | Kenichi Maeda, Computer Science, University of Massachusetts Boston |
| 2023–2026 | Edward Gaibor, Computer Science, University of Massachusetts Boston |
| 2022–2024 | Ryan Zurrin, Computer Science, University of Massachusetts Boston |
| 2023 | Amanda Kwong, Conputer Science and Honors Thesis, University of Massachusetts Boston |
| 2022–2023 | Josh Kotler, Computer Science, University of Massachusetts Boston |
| 2022–present | Akshata Tiwari, Pre-college student at Aliso Niguel High School |
| 2022 | Kendrick Kheav, Biochemistry, University of Massachusetts Boston |
| 2022 | Nikol Vladinska, Computer Science and Honors Thesis, University of Massachusetts Boston |
| 2021–2022 | Jay Burkhardt, Computer Science, University of Massachusetts Boston |
| 2021 | Isabelle Lara, Biology, University of Massachusetts Boston |
| 2021 | Patricia Somera, Biology, University of Massachusetts Boston |
| 2021 | Bella Baidak, Computer Science, University of Massachusetts Boston |
| 2019–2021 | Yahiya Hussain, Computer Science, University of Massachusetts Boston |
| 2019–2021 | Nandinii Yeleswarapu, Computer Science and Honors Thesis, University of Massachusetts Boston |
| 2019–2020 | Safwa Ali, Engineering, University of Massachusetts Boston |
| 2019–2020 | Huda Irshad, Engineering, University of Massachusetts Boston |
| 2018–2020 | Ian Svetkey, Pre-college student at Harvard University |
| 2016 | Eagon Meng, Computer Science, Harvard University |
| 2016 | Omar Shaikh, (Remote-) Intern at Harvard University |
| 2015–2017 | John Hoffer, Computer Science, Harvard University |

## Mentoring (continued)

| | |
|---|---|
| 2015 | William Zhang, Pre-college student at Harvard University |
| 2013 | Jay Andrew Robinson, Intern at Boston Children's Hospital (co-mentored) |
| 2013 | Emily Seibring, Intern at Boston Children's Hospital (co-mentored) |

## Service and Outreach

### Departmental Level

| | |
|---|---|
| 2025–2026 | Chair of the Faculty Search Committee |
| 2023–present | Member of the Student Activities Committee |
| 2022–present | Member of the Broadening Participation in Computing Task Force |
| 2022–present | Faculty Advisor for the Computer Science Club |
| 2020–present | Organizer of Events and Maintainer of the Discord Server for CS+IT Students |
| 2020 | Member of the Paul English Scholarship Committee |
| 2019–present | Member of the Outreach and Publicity Committee |
| 2019–present | Member of the Student Recruitment Committee |
| 2019–2020 | Organizer of bi-weekly social events for IT and CS students |

### College and University Level

| | |
|---|---|
| 2026 | Faculty Marshall at Graduate Commencement |
| 2026 | Member of the Manning Teaching Prize Selection Committee |
| 2026 | Member of the CSM Early Career Research Excellence Award Selection Committee |
| 2026 | Member of the CIT Award Selection Committee |
| 2025–present | Faculty Lead for the Student Experience Project and Community of Practice |
| 2025–present | Board Member of the Center for Innovative Teaching |
| 2025 | Member of the Research Computing Hiring Committee |
| 2024–present | Faculty Advisor for the UMass Boston Chapter of the National Society of Black Engineers (NSBE) |
| 2024 | Member of the Research Computing Hiring Committee |
| 2024 | Chair of the Poster Judging Committee for the CSM Student Success Showcase |
| 2024 | Member of the Faculty Search Committee for Physics-based Machine Learning |
| 2024 | Member of the CSM Student Success Showcase Organization Committee |
| 2023 | Judge for the CSM Student Success Showcase Poster Awards |
| 2023 | Member of the Research Computing Hiring Committee |

## Service and Outreach (continued)

| | |
|---|---|
| 2022 | Invited Researcher for the UMass President's Office NSF Type 2 Engine proposal for Equitable Health |
| 2022 | Member of the Research Computing Hiring Committee |
| 2022 | Member of the Data Science Faculty Search Committee |
| 2022 | Member of the Joint Discipline & Grievance Committee |
| 2021 | Panelist for the Brain Trusts in AI/Robotics/Data Science Initiative from the UMass President's Office |
| 2020–present | Member of the Research Computing Advisory Committee |
| 2020–2021 | STEM Educational Excellence (STEM-EdX) Fellow |
| 2020 | Member of the Data Science Faculty Search Committee |

### Professional Activities

| | |
|---|---|
| 2026 | Chair for the Tissue Section at the VIZBI Conference in Cambridge, UK |
| 2026 | Tenure Evaluator for the University of Southern California |
| 2026 | Reviewer for the Biobank program at the Massachusetts Life Sciences |
| 2025 | Reviewer for the AP Computer Science exam |
| 2025 | Co-Organizer of the ACM-W MAGICC conference at Amherst College |
| 2025 | Reviewer for the Health Equity Accelerator at the Massachusetts Life Sciences |
| 2024 | White House Summit on STEMM Equity and Excellence, Leveraging Diverse Minds in R&D Discussion |
| 2024 | Host and Organizer of the first Boston Brainhack event at UMass Boston |
| 2023 | Trained Instructor for the National Research Mentoring Network (NRMN) |
| 2023 | Reviewer for the MICCAI Grand Challenge for Aorta Segmentation |
| 2023 | Public Interest Technology (PIT) New England Member |
| 2022 | National Science Foundation Reviewer and Panelist for SBIR Grants |
| 2022 | Program Committee member at the IEEE Visualization conference |
| 2022 | Topic Editor: Open Source for Open Science, Frontiers in Neuroinformatics |
| 2021 | Program Committee member at the IEEE Visualization conference |
| 2021 | Faculty Mentor at the MGH Neuroimaging 2021 Virtual Symposium |
| 2021 | Organizer of the Chart Question Answering Workshop at CVPR 2021 in collaboration with Harvard, Columbia, Northwestern, and UMass Amherst |
| 2021 | National Science Foundation Reviewer and Panelist for SBIR Grants |
| 2020 | Program Committee member for short papers at the IEEE Visualization conference |
| 2018–present | Reviewer for *Manning Publications* |

# Service and Outreach (continued)

| | |
|---|---|
| 2016–present | Reviewer for Frontiers in Neuroinformatics, ISMRM, Neuroinformatics, Frontiers in Neural Circuits, ACM SIGCHI, IEEE CVPR, ECCV, IEEE Visualization / Transactions on Visualization and Computer Graphics, IEEE Access, MDPI Applied Sciences, Nature Communications Biology, Scientific Reports, Transactions on Pattern Analysis and Machine Intelligence, Nature, Computer & Graphics, NeurIPS, IEEE Computer Graphics and Applications, BELONGING Journal, Journal of Neuroimaging, IEEE ISBI |
| 2013 | Technical Reviewer for *Matsuda and Lea: WebGL Programming Guide, Addison-Wesley* |

## Community Service

| | |
|---|---|
| 2025 | CS Girlies Hackathon Judge |
| 2025–present | Public Interest Technology New England (PIT-NE) Leadership Committee |
| 2024–present | Public Interest Technology New England (PIT-NE) Ambassador for UMass Boston |
| 2023 | Volunteer for the Petey Greene Program to tutor incarcerated people |
| 2022 | Member of the Principal Search Committee for the Putnam Ave Upper School in Cambridge |
| 2019–2020 | Advisor for the AP Data Science Curriculum in Cambridge Public Schools |
| 2018–2019 | Head Coach for Cambridge Youth Soccer |
| 2018 | Volunteer+Presentation Facilitator at the Cambridge 8th Grade Science & Engineering Showcase |
| 2007–2010 | President of the Student Computer Club at Heilbronn University, StuWoNet e.V. |
| 2007–2009 | Voluntary Project Lead of RANDI2, a randomization software for clinical trials at the German Cancer Research Center (DKFZ), coordinating 15+ developers |
| 1997–1999 | Vice-President of The German Computer Freaks, a National Cyber Security Club |

My Erdős Number is 3.